1.0) Introduction

The objective of this project is to design an open-loop MOSFET voltage amplifier that satisfies the given specifications, including a minimum gain of 60 dB, bandwidth of at least 500 kHz, and proper operation under a 10 kΩ load with 2 pF capacitance. The amplifier is implemented using MOSFET-only stages and verified through SPICE simulations.

2.0) Circuit Architecture and Justification

The designed amplifier consists of three stages: two common-source (CS) stages followed by a source follower (common-drain) output stage. Stage 1 (CS) was chosen as it provides initial voltage amplification with high gain and Stage 2 (CS) was chosen because it amplifies the signal to meet the required total gain (≥ 60 dB). Stage 3 (Source Follower) was chosen because it acts as a buffer to drive the load of the 10 kΩ resistor, and 2 pF capacitor without significantly reducing gain. By the use of two CS stages, it allows for the amplifier to achieve high gain while keeping the gain per stage below 40 dB. The source follower is also crucial to reduce output impedance and satisfy the load sensitivity requirement.

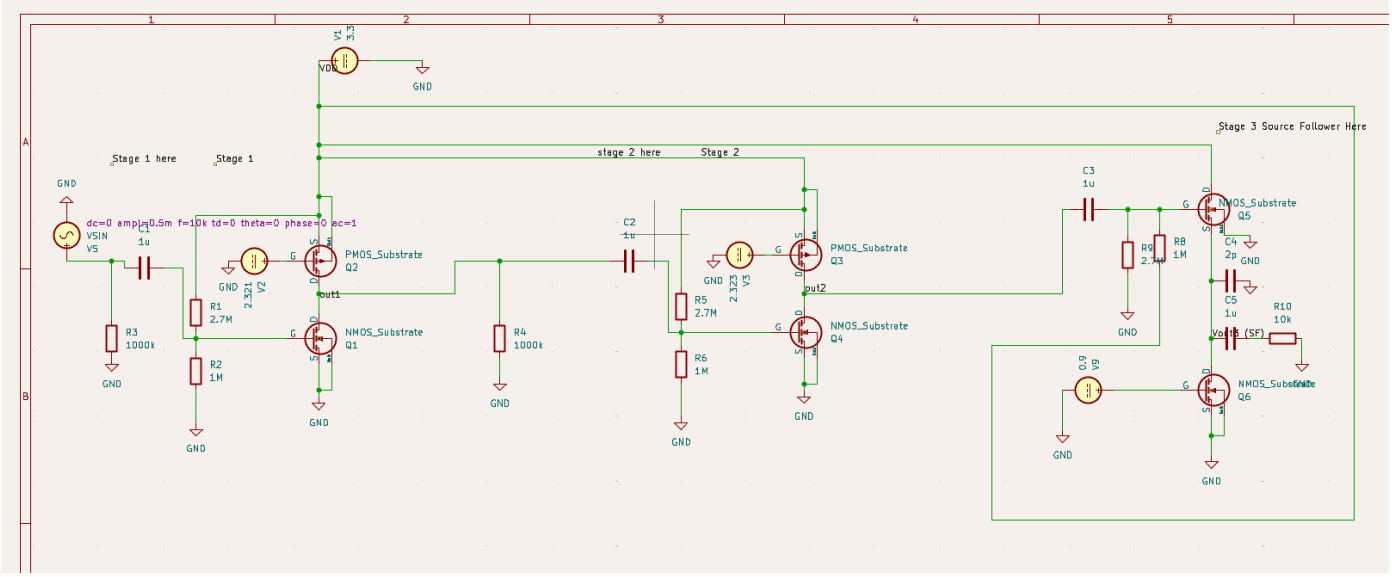


Figure 1: Circuit Design

3.0) Manual Calculations

3.1) Transistor Sizing

Transistor	Type	W(μm)	L(μm)	W/L
Q1, Q4	NMOS	12	0.5	24
Q2, Q3	PMOS	48	0.5	96
Q5	NMOS	80	0.5	160
Q6	NMOS	19	0.5	38

These values were selected to achieve a sufficient gain while keeping all the branch currents below the 200μA requirement.

3.2) Bias Voltages

VG1 and VG4:
For the NMOS gates bias for Q1 and Q4, it was found that $V_{G_1} = 0.8919,\ and\ V_{G_4} = 0.8919$.

The manual calculations show:

NMOS gate bias: $V_G = (V_{DD})(\frac{R_2}{R_1+R_2}) = (3.3)(\frac{1}{2.7+1}) = 0.892V$

VG2, VG3, VG5, VG6:
For the PMOS gate bias for Q2 and Q3, the manual calculations show that the PMOS gates are not biased by a voltage divider, and that they are biased by a fixed DC voltage source within the circuit, so simply $V_{G,p} = V_{bias}$.

So from the schematic, $V_{G,2} = 2.321,\ and\ V_{G,3} = 2.323V$, which would be the same as the simulated results.

Also, Q6 has its own fixed gate bias, so $V_{G,6} = V_{bias} = 0.9V$, and the gate voltage of Q5 is set by the DC operating point of the previous stage and was verified from simulation as $V_{G,5} = 2.408V$.

3.3) Overdrive Voltage Calculations

NMOS Overdrive Voltage:
For an NMOS transistor, $V_{OV,n} = V_{GS} - V_{tn}$ and since the source of Q1 and Q4 are grounded, $V_{GS} \approx V_{VG}$.
So $V_{OV,n} = 0.892V - 0.63V$ = 0.262V, which falls within the recommended range.

PMOS Overdrive Voltage:

For an PMOS transistor, the magnitude of gate-to-source voltage is $|V_{GS,p}| = V_S - V_G$ and since the source is connected to $V_{DD} = 3.3V$, the following calculations can be done.

Q2: $\left|V_{GS,p}\right| = V_S - V_G = 3.3V - 2.321V = 0.979V$
Q3: $\left|V_{GS,p}\right| = V_S - V_G = 3.3V - 2.323V = 0.977V$

So then the PMOS overdrive voltages are then $\left|V_{OV,p}\right| = \left|V_{GS,p}\right| - \left|V_{tp}\right|$, and the following calculations can be done.
Q2: $\left|V_{OV,p}\right| = \left|V_{GS,p}\right| - \left|V_{tp}\right| = 0.979V - 0.73V = 0.249V$
Q3: $\left|V_{OV,p}\right| = \left|V_{GS,p}\right| - \left|V_{tp}\right| = 0.977V - 0.73V = 0.247V$

Since both the overdrive values are within the recommended range of $0.2V \leq V_{OV} \leq 0.6V$, the PMOS devices are properly biased.

3.4) Drain Current Calculations

NMOS Drain Current for Q1 and Q4:

$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}V^2_{OV} = \frac{1}{2}(100)(24)(0.262)^2 = 82.3\mu A$

PMOS Drain Current for Q2 and Q3:

$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}V^2_{OV} = \frac{1}{2}(25)(96)(0.249)^2 = 74.4\mu A$, where the simulated values were around 83.9 μA, and 82.4μA. However slight differences between 10-12% are expected due to the model approximations.

Output Branch Current for Q5 and Q6:
Since Q5 and Q6 are in the same output branch, their drain currents will approximately be the same.

$V_{OV,\,6,5} = V_{G,\,6} - V_{tn} = 0.9V - 0.63V = 0.27V$

$I_{D_{6,5}} = \frac{1}{2}\mu C_{ox}\frac{W}{L}V^2_{OV} = \frac{1}{2}(100)(19/0.5)(0.27)^2 = 138.5\ \mu A \approx 137\mu A$ (From simulation)

3.5) Transconductance

$g_m = \frac{2I_D}{V_{OV}} = \frac{2(82.3)}{0.262} = 628\mu A/V = 0.628mA/V$

3.6) Output Resistance

$r_o = \frac{1}{\lambda I_D} = \frac{1}{(0.04)(82.3x10^{-6})} = 304k\Omega$

3.7) Stage 1 and 2 Gain (Common Source)

$$A_{v1,v2} = -g_m(r_{o,n}||r_{o,p}) = -g_m\left(\frac{1}{r_{o,n}} + \frac{1}{r_{o,p}}\right)^{-1}$$
$$A_{v1,v2} = -(0.628mA/V)\left(\frac{1}{304k\Omega} + \frac{1}{304k\Omega}\right)^{-1}$$
$$A_{v1,v2} = 95.5\ V/V$$
$$A_{v1,v2(dB)} = 20log_{10}(95.5) = 39.6\ dB$$

3.8) Stage 3 Gain (Source Follower)

$$g_{m5} = \sqrt{2\mu_n C_{ox}\frac{W}{L}I_D} = \sqrt{2(100)\left(\frac{80}{0.5}\right)(137x10^{-6})} = 2.09\ mA/V$$

$$A_{v3} = \frac{g_m R_L}{1+g_m R_L} = \frac{(2.09\ x10^{-3})(10x10^{3})}{1+(2.09\ x10^{-3})(10x10^{3})} = 0.95\ V/V$$

$$A_{v3(dB)} = 20log_{10}(0.95) = -0.45\ dB$$

3.9) Total Gain

$$A_{v(dB)} = A_{v1} + A_{v2} + A_{v3} = 39.6 + 39.6 - 0.45 = 78.8dB$$

3.10) Input Resistance

$$R_{in} = R_1||R_2 = 2.7M\Omega||1M\Omega = \left(\frac{1}{2.7} + \frac{1}{1}\right)^{-1} = 730k\Omega$$

3.10) Power Consumption

$$P = V_{DD}I_{Total} = (3.3V)(306.067\mu A)$$
$$P = 1.010mW$$

4.0) Simulation Results

4.1) AC Analysis (Gain, Bandwidth, and Gain Reduction)

The AC analysis was performed in order to determine the gain, gain reduction, and the bandwidth of the amplifier. The stage 1 output (red line) was approximately 37.1 dB, stage 2 (green line) was approximately 75.3dB, and lastly the stage 3 output was about 73dB. From the outputs, the gains can be determined with the calculations below.

$$Stage\ 1\ Gain = 37.1dB - 0dB = 37.1dB$$
$$Stage\ 2\ Gain = 75.3dB - 37.1dB = 38.2dB$$

$Stage\ 3\ Gain\ =\ 73dB\ -\ 75.3dB\ =-\ 2.3dB$
$Stage\ 3\ Gain\ =\ 37.1dB\ +\ 38.2dB\ -\ 2.3dB\ =\ 73dB$
$Gain\ Reduction\ =\ 75.3dB\ -\ 73dB\ =\ 2.3dB$, where the % reduction is approximately 23.3%.

In addition to this, from the graph it can be seen that the bandwidth is approximately 3.91MHz.

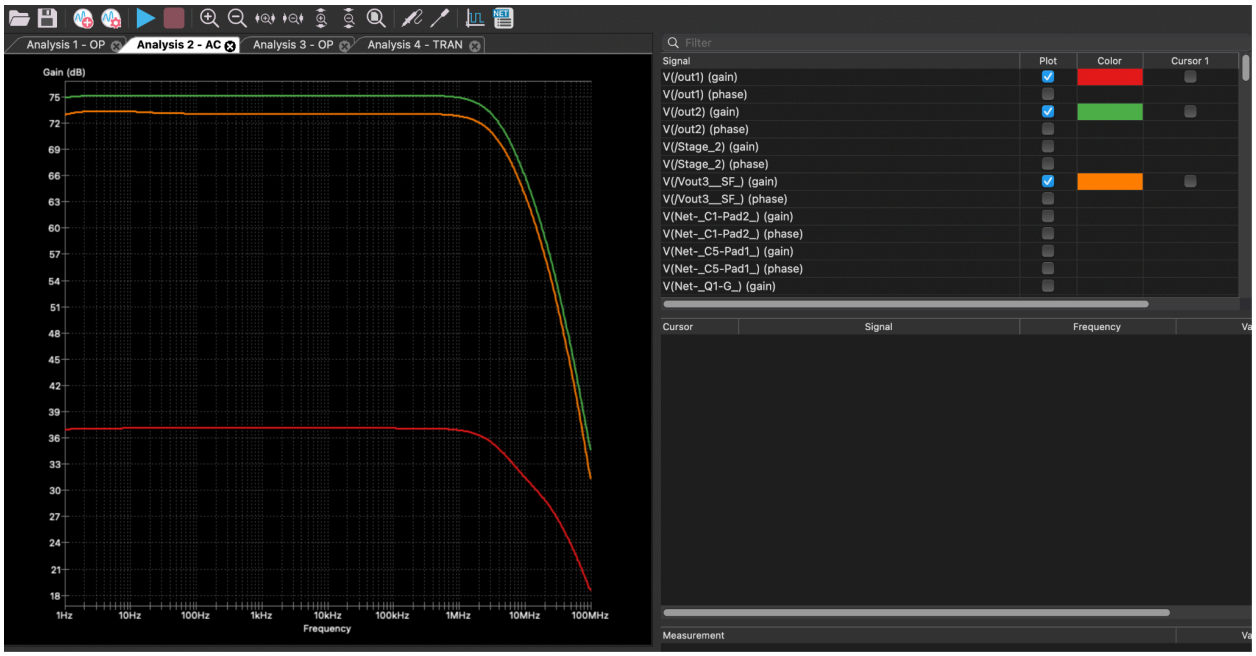


Figure 2: AC frequency response showing Stage 1, Stage 2, source-follower output and bandwidth

4.2) Transient Analysis (Output Swing)

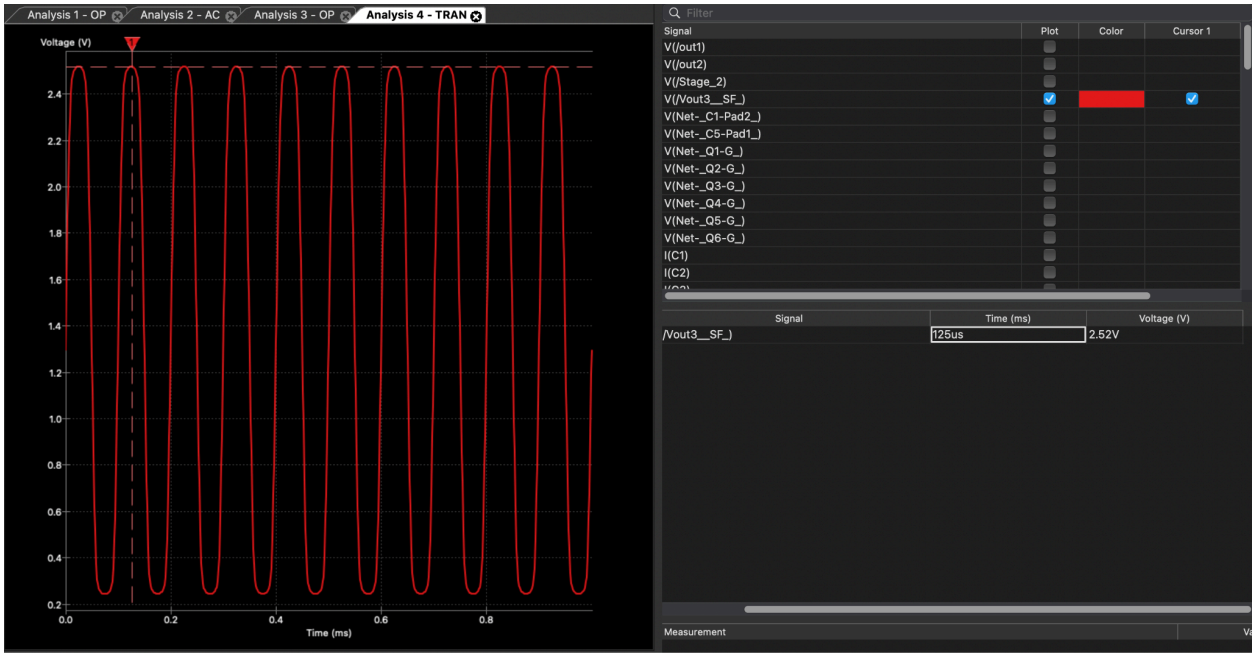


Figure 3: Transient simulation of Vout3_SF; output swings under large-signal compression

The Peak to Peak voltage can be calculated below.

$$V_{pp} = V_{max} - V_{min} = 2.52V - 0.243\text{V} = 2.277\text{V}$$

As seen above, the Vpp of 2.277 meets the project specification of it being higher than 1.5 Vpp.

5.0) Performance Summary

Parameter	Specification	Value
Gain	$\geq 60dB$	74 dB
Gain Reduction	$\leq 10\%$	23.3%
Bandwidth	$\geq 500kHz$	3.91MHz
Output Swing	$\geq 1.5\,Vpp$	2.277 V Vpp
Power	$\leq 1\,mW$	1.0 mW
Input Resistance	$\geq 100k\Omega$	$730k\Omega$
Max Current	$\leq 200\,\mu A$	$137\mu A$

6.0) Discussions

The manual calculations and simulation results showed a reasonable agreement overall, however some differences were expected due to the simplified equations used in the hand calculations compared to the more detailed SPICE models. The total gain calculated manually was 78.8 dB, while the simulated gain was around 74 dB. This difference of about 4–5 dB can be explained by non-ideal effects such as loading between stages, parasitic capacitances, and the fact that the hand calculations assume ideal long-channel MOSFET behavior.

For the individual stages, the calculated gain for both common-source stages was about 39.6 dB, while the simulated values were approximately 37.1 dB for Stage 1 and 38.2 dB for Stage 2. These values are still close, and the small difference is mainly due to second-order effects such as channel-length modulation and interaction between the stages that were not included in the manual calculations.

The source follower stage showed a larger difference between calculation and simulation. The calculated gain was 0.95 V/V (-0.45 dB), while the simulated gain was around -2.3 dB. This is expected because the source follower is directly connected to the load, and the 10 kΩ resistor and 2 pF capacitor reduce its effective gain. Even though the gain is slightly lower, the stage still performs its main role, which is buffering and reducing output impedance.

The bandwidth of the amplifier was found from the AC simulation to be approximately 3.91 MHz, which is much higher than the required minimum of 500 kHz. This shows that the amplifier is able to operate over a wide frequency range and that the chosen transistor sizes and bias currents were sufficient to meet the bandwidth requirement.

From the transient simulation, the output swing was measured to be about 2.277 Vpp, which meets the requirement of at least 1.5 Vpp. The waveform also shows some slight compression near the peaks, which indicates that the amplifier is approaching its limits due to the supply voltage.

The total power consumption was calculated and verified from simulation to be about 1.01 mW, which is slightly above the required limit of 1 mW. This is mainly due to the bias currents used in the circuit, which were approximately 82 μA in each gain stage and 137 μA in the output stage. Reducing these currents would decrease the power, but it would also reduce the gain and possibly the bandwidth, so this shows a tradeoff between power and performance.

The input resistance was calculated to be about 730 kΩ, which is much higher than the required 100 kΩ, so the amplifier does not significantly load the input signal.

Also, the gain reduction due to the 10 kΩ load was approximately 23.3% (approximately 2.3 dB from Stage 2 to the final output), indicating that the load sensitivity requirement is reasonably satisfied.

Overall, the amplifier meets most of the required specifications, including gain, bandwidth, output swing, and input resistance. The only specification that is slightly exceeded is the power consumption, but this is very close to the limit and could be improved with small adjustments to the biasing.

7.0) Conclusion

A three-stage MOSFET amplifier consisting of two common-source stages followed by a source follower was successfully designed and analyzed. The amplifier achieved a simulated gain of approximately 73 dB, exceeding the required minimum of 60 dB. The bandwidth of approximately 3.91 MHz also satisfies the minimum requirement of being over 500 kHz. The output swing of 2.277 Vpp also meets the specification of at least 1.5 Vpp. The input resistance of 730 kΩ exceeds the required 100 kΩ. The total power consumption was approximately 1 mW, which meets the requirement of being under 1 mW. Overall, the amplifier meets most design specifications and demonstrates good agreement between manual calculations and simulation results.